

PIROVSKOYE, Russian Federation

WMO: 293630

Lat: 57.6312N

Lon: 92.2714E

Elev: 588

StdP: 14.39

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.0	-37.3	-47.7	0.3	-43.0	-40.9	0.5	-35.9	16.6	16.5	15.0	14.5	2.0	60	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	84.4	66.1	81.0	64.7	77.7	63.6	68.6	78.8	67.1	77.0	65.4	74.6	5.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	95.9	72.7	63.6	89.9	71.0	61.8	84.4	69.7	33.3	78.8	32.0	77.1	30.7	74.9	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.6	12.0	DB	-50.9	88.1	4.5	2.9	-54.2	90.1	-56.8	91.8	-59.3	93.4	-62.6	95.5	(4)
(5)			WB	-48.8	71.2	2.9	2.2	-50.9	72.8	-52.6	74.1	-54.2	75.4	-56.2	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.1	-5.7	4.4	19.1	34.9	46.9	60.6	63.7	59.3	45.6	30.8	9.3	2.7		
	DBStd	26.67	18.26	14.71	13.37	8.72	9.88	8.11	6.25	6.73	7.71	10.36	17.06	17.98		
	HDD50	8088	1728	1276	957	458	185	16	0	9	176	595	1220	1468		
	HDD65	12500	2193	1696	1422	903	568	172	100	201	583	1059	1670	1933		
	CDD50	1191	0	0	0	4	87	335	425	296	43	1	0	0		
	CDD65	129	0	0	0	0	5	41	60	23	0	0	0	0		
	CDH74	1583	0	0	0	2	73	624	626	252	6	0	0	0		
	CDH80	363	0	0	0	0	12	155	150	46	0	0	0	0		
	WSAvg	5.4	5.3	5.4	5.8	6.6	6.4	4.9	4.2	4.3	4.8	5.5	6.0	5.9		
	PrecAvg	24.7	1.5	1.1	1.2	1.6	2.0	2.5	2.4	2.7	2.6	2.6	2.6	2.0		
(15) Precipitation	PrecMax	31.2	3.1	1.9	2.3	2.6	4.0	4.4	4.9	4.2	4.5	3.8	4.2	3.4		
	PrecMin	18.7	0.1	0.5	0.2	0.6	0.8	0.3	0.6	1.3	0.6	0.8	1.0	0.6		
	PrecStd	3.2	0.6	0.4	0.5	0.5	0.8	0.9	1.0	0.8	0.9	0.7	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	34.2	48.2	66.4	82.3	88.5	87.3	84.7	73.9	60.8	44.5	33.7		
		MCWB	28.8	29.4	38.2	50.6	59.0	65.9	69.0	68.4	58.6	47.9	41.3	31.7		
	2%	DB	27.1	30.9	42.8	59.7	75.2	84.6	84.5	80.3	67.5	53.1	38.9	30.5		
		MCWB	25.2	27.9	35.0	46.4	57.5	64.8	67.3	66.3	56.4	44.5	35.6	29.3		
	5%	DB	23.1	28.4	39.7	53.9	69.9	80.8	80.9	76.3	63.2	48.1	35.6	27.8		
		MCWB	21.7	26.1	33.4	43.0	54.4	63.2	66.2	64.3	54.4	42.8	33.1	26.7		
	10%	DB	18.9	25.3	36.5	49.5	65.1	77.3	77.3	72.6	58.6	44.1	31.7	24.6		
		MCWB	17.6	23.3	31.7	40.7	52.2	62.2	64.5	62.7	51.9	40.5	30.4	23.5		
	0.4%	WB	29.3	30.7	40.1	51.5	62.3	69.1	72.9	70.9	60.3	49.2	42.1	32.3		
		MCDB	31.2	32.9	47.0	61.6	78.4	81.6	85.3	79.2	68.6	58.0	45.3	33.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	25.2	28.1	36.3	47.4	58.6	67.1	69.2	67.8	57.9	45.7	36.4	29.5		
		MCDB	26.7	30.4	41.3	57.4	71.6	78.9	79.4	77.4	65.2	51.6	38.6	30.5		
	5%	WB	21.8	26.3	34.0	44.1	55.7	65.6	67.8	66.0	55.5	43.2	33.2	26.6		
		MCDB	23.2	28.7	38.4	52.6	67.9	77.0	77.4	74.6	62.8	47.6	35.2	27.6		
	10%	WB	17.7	23.2	32.0	41.1	53.1	63.9	66.0	63.8	52.5	41.0	30.3	23.5		
		MCDB	19.1	25.4	36.4	48.8	64.0	74.3	74.7	70.9	57.6	44.8	32.0	24.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	18.2	19.7	17.1	19.5	21.5	20.0	19.0	16.1	12.5	12.8	13.8		
		MCDBR	14.4	15.4	20.0	24.9	27.7	29.1	26.2	25.0	25.1	18.8	11.1	13.4		
		MCWBR	13.6	13.4	14.3	14.5	13.9	11.9	11.5	12.1	14.3	12.4	9.6	12.5		
		MCDBR	14.0	14.3	16.8	22.6	24.6	23.4	21.6	21.3	22.3	17.0	11.6	12.9		
	5% WB	MCWBR	13.4	12.7	12.6	13.9	13.3	10.7	10.6	11.0	13.5	12.0	10.5	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.317	0.356	0.386	0.403	0.412	0.434	0.408	0.365	0.342	0.306	0.278		
	taud		1.974	1.972	1.944	2.068	2.221	2.281	2.227	2.278	2.361	2.244	2.021	1.930		
	Ebn at Noon		165	215	242	258	263	262	252	251	245	211	154	125		
	Edh at Noon		20	31	42	44	40	39	40	36	28	24	19	15		
(44) All-Sky Solar Radiation	RadAvg		208	509	935	1401	1586	1892	1738	1345	810	397	194	118		
	RadStd		28	41	69	93	166	179	120	113	110	37	36	23		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page